

ASSIGNMENT GROUP WORK

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A. Introduction

In the contemporary landscape of Educational Technology (EdTech), the integration of Artificial Intelligence (AI) into self-directed learning has transitioned from a luxury to an absolute necessity. Traditional mobile learning applications often provide static databases or flat-list flat-list course structures, leaving students without interactive, real-time guidance when tackling complex STEM (Science, Technology, Engineering, and Mathematics) problems outside of school hours. To address this critical gap, our development group (BrightPath Learning) initiated the "AI Study Mentor" project—a dynamic mobile application designed to simulate a 24/7 personal tutor for high school and university students.

This comprehensive software design document chronicles the end-to-end architectural planning and project management of the "AI Study Mentor" Minimum Viable Product (MVP) under a strict 14-week academic deadline.

The report is systematically structured to reflect industry-standard software engineering practices. It begins by justifying the selection of the Agile methodology and specific development tools (Android Studio, Git, Jira) required to manage a 5-member development team. Subsequently, it delves into Object-Oriented Analysis and Design (OOAD), utilizing standard UML visualizations (Use Case and Class Diagrams) to establish a robust, modular system architecture. Furthermore, this document emphasizes risk mitigation through a rigorous Quality Assurance Test Plan, capturing edge cases and API timeout handling. Finally, by conducting comprehensive peer reviews and user surveys, the report critically evaluates system trade-offs—specifically balancing the need for rapid AI inference (via Grok Cloud API) against zero-budget constraints and data security—laying a scalable foundation for future technological enhancements.

B. Content

II. Activity 2: Software Development Tools and Techniques

2.1. Software Development Life Cycle (SDLC) Models

(Sommerville, 2015)The software development lifecycle (SDLC) is a structured and iterative methodology used by development teams to build, deliver and maintain high-quality and cost-effective software systems.

The SDLC breaks down [software development](#) into distinct, repeatable, interdependent phases. Each phase of the SDLC has its own objectives and deliverables that guide the next phase. Taken together, the phases of the SDLC form a roadmap that helps development teams create software that meets stakeholder needs, project requirements and customer expectations.

There are various SDLC models, and each approaches the phases of the SDLC in its own way. In some models, such as the waterfall model, the phases are completed sequentially. In other, more iterative processes, such as agile, phases can be worked on in parallel.

A. Agile Methodology

Definition: Agile is an iterative and incremental approach that breaks the project into short development cycles called sprints (typically 1-4 weeks). It emphasizes continuous feedback, team collaboration, and delivering functional software early.



Figure 1: Agile Methodology

Strengths: Highly adaptable to changing requirements. It allows continuous testing and feedback, which is beneficial when developing complex features like AI answer generation

Weaknesses: Requires intense communication and can lead to scope creep if requirements are not well-managed. Documentation might be less prioritized.

B. Waterfall Methodology

Definition: Waterfall is a linear, sequential SDLC model where each phase (Requirements, Design, Implementation, Testing, Deployment, Maintenance) must be fully completed before the next one begins.

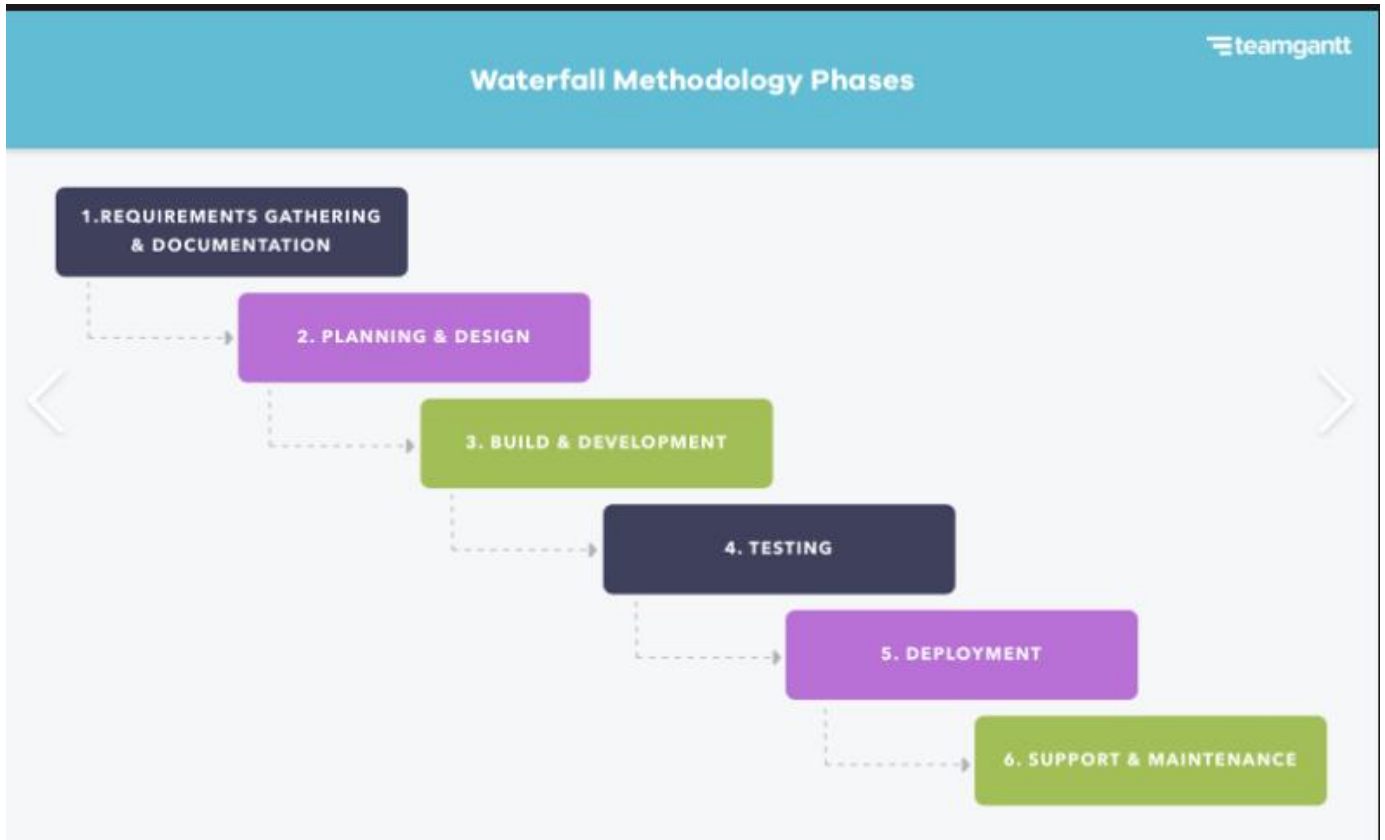


Figure 2: Waterfall Methodology

Strengths: Extremely structured and easy to manage. Extensive documentation is created at every stage, providing clear milestones.

Weaknesses: Inflexible to changes. If the BrightPath Learning team discovers integration issues with AI APIs late in the testing phase, revisiting the design is highly costly and time-consuming.

C. Spiral Methodology

Definition: The Spiral model combines the iterative nature of Agile with the systematic aspects of Waterfall, placing a heavy emphasis on continuous risk analysis.

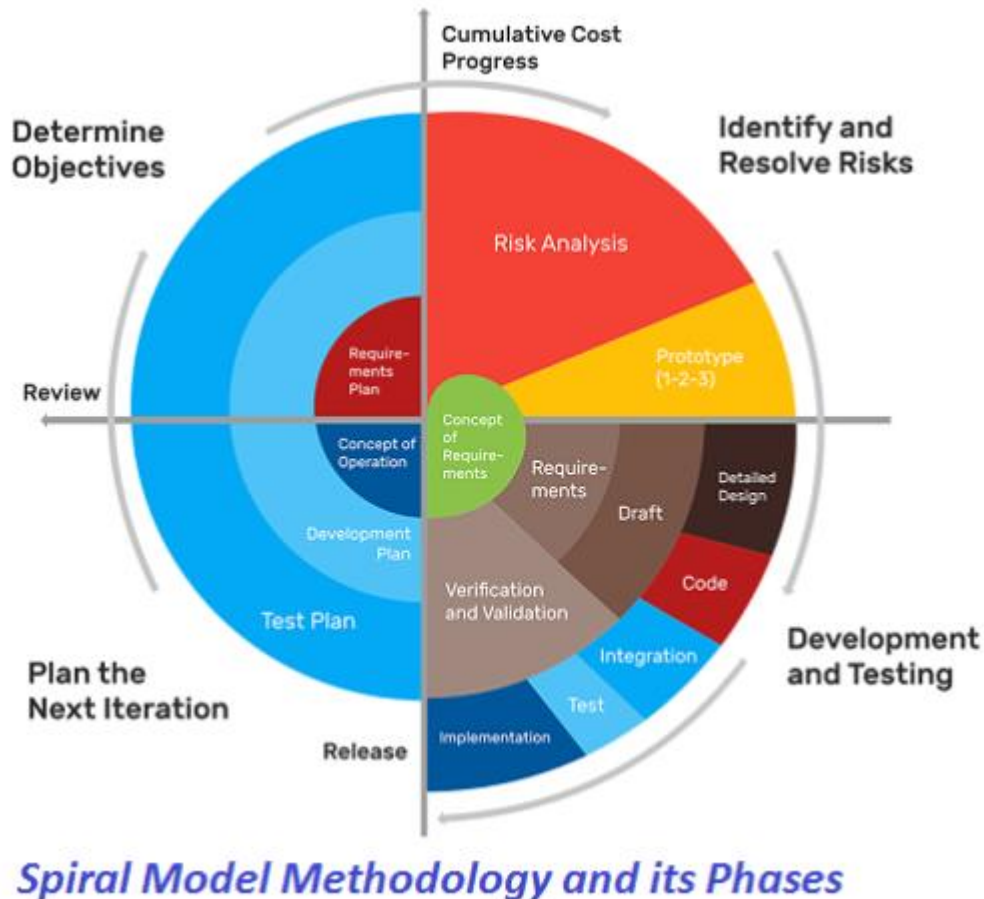


Figure 3: Spiral Methodology

Strengths: Excellent for identifying and mitigating risks early, especially concerning budget constraints or AI cost management.

Weaknesses: It is complex and can be overly expensive and time-consuming for small projects with tight deadlines.

2.2. Development Tools

Development tools are software applications that assist developers in creating, debugging, maintaining, and managing software projects.

A. Integrated Development Environment (IDE): Android Studio

Definition: (Google, 2023) Android Studio is the official IDE for Google's Android operating system, providing a unified environment for coding, testing, and debugging.

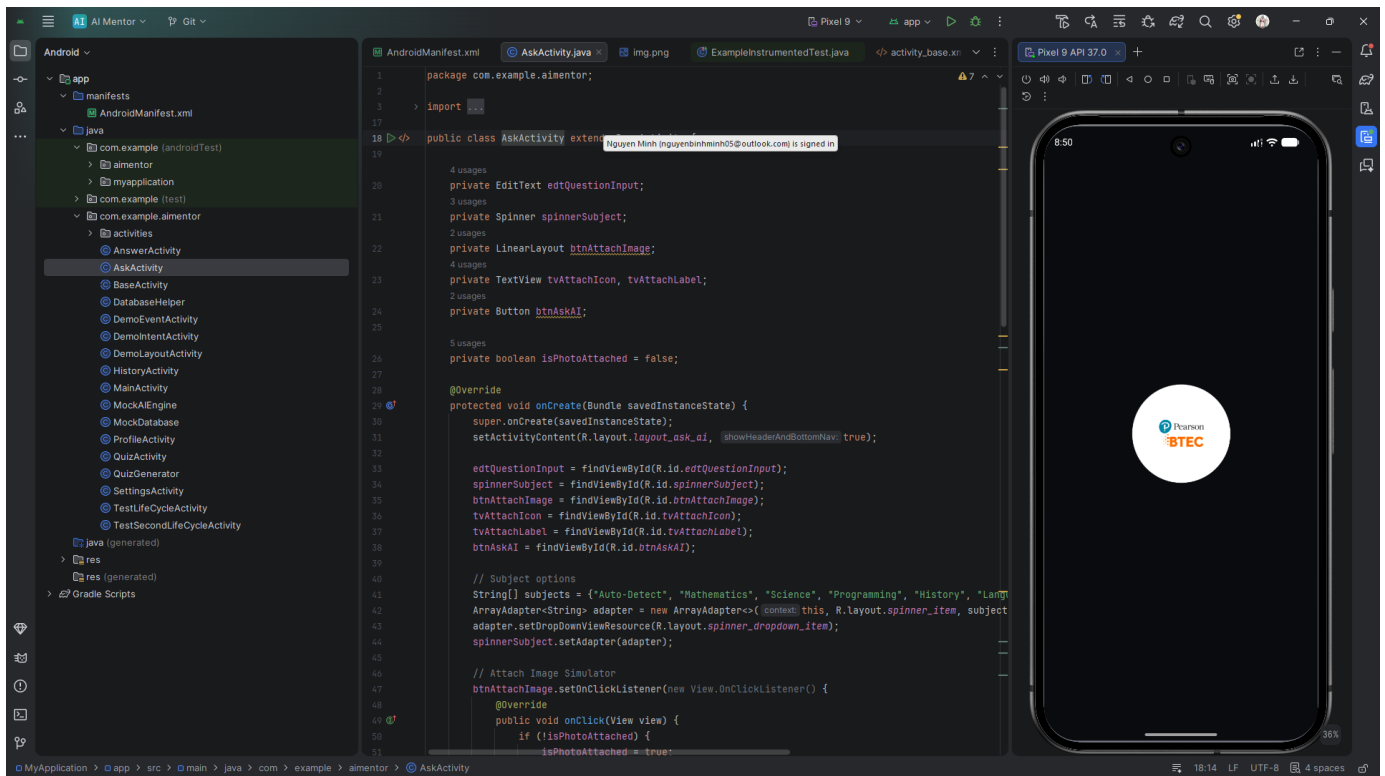


Figure 4: Android Studio programming interface

Strengths: Offers robust code editing, intelligent code completion, and built-in emulators. It provides extensive libraries and frameworks suitable for junior developers.

Weaknesses: Consumes significant system resources (RAM and CPU) and has a steep learning curve for absolute beginners.

B. Project Management Tool: Jira

Definition: (Atlassian, 2023) Jira is a proprietary issue tracking and agile project management product developed by Atlassian.



Figure 5: Jira

Strengths: Excellent for tracking sprints, assigning tasks, and managing bugs, keeping the development team organized.

Weaknesses: The interface can be overwhelming, and setting up workflows requires initial effort.

C. Version Control System: Git & GitHub

Definition: (Software Freedom Conservancy, 2023) Git is a distributed version control system, while GitHub is a cloud-based hosting service that lets developers manage Git repositories.



Figure 6: Git



Figure 7: Github

Strengths: Facilitates seamless team collaboration, tracks all code changes, and allows branching for feature development without breaking the main application.

Weaknesses: Requires developers to understand command-line operations and merging conflicts can occasionally be complex.

D. Artificial Intelligence Engine: Grok Cloud API

Definition: Grok Cloud provides developer API access to the Grok Large Language Model (LLM), an advanced AI system designed for real-time natural language processing, logical reasoning, and complex problem-solving.

Strengths: It offers highly optimized response latency and highly cost-effective API usage rates, making it the perfect engine to power a fast-paced, budget-constrained educational MVP like "AI Study Mentor".

Weaknesses: As a rapidly emerging AI ecosystem, its third-party Android wrapper libraries and community troubleshooting forums are still evolving compared to older legacy models.



Grok API

xAI Grok API Resources

Figure 8Grok Cloud AP:

2.3. Justification for Selected Methodology and Tools

Android Studio (Integrated Development Environment)

Why it was selected: Android Studio is the official IDE for Android development. Unlike lightweight text editors like VS Code, which require configuring multiple plugins to compile mobile apps, Android Studio provides a comprehensive, out-of-the-box environment specifically tailored for native mobile performance.

How it specifically benefits "AI Study Mentor":

Handling Hardware Features: Our app requires students to take photos of their homework (e.g., math problems). Android Studio provides native libraries (like CameraX) and easy permission management, making it straightforward to implement the image capture feature securely.

UI/UX Design: The built-in XML Layout Editor allows the team to visually drag-and-drop UI components. This accelerates the creation of the chat interface where students interact with the AI, saving valuable time during the strict 14-week timeline.

Performance Monitoring: When the app receives long, complex step-by-step explanations from the AI API, it consumes significant memory. Android Studio's built-in **Android Profiler** helps the team monitor CPU and RAM usage in real-time, preventing the app from crashing on low-end student devices.

Git & GitHub (Version Control System)

Why it was selected: Git is a distributed version control system that tracks every line of code modified, while GitHub acts as a centralized cloud repository. It was chosen over centralized systems (like SVN) or sharing code via Google Drive because it natively supports concurrent development.

How it specifically benefits "AI Study Mentor":

Parallel Development for a 5-Member Team: With five members working simultaneously (Nguyen Binh Minh, Nguyen Huu Duc Anh, Nguyen Dinh Thanh, Tran The Vinh, Nguyen Ngoc Minh), merge conflicts are a major risk. Git allows each member to create a separate "branch" (e.g., feature/login, feature/ai-integration, feature/camera-upload).

Safe AI Experimentation: Integrating AI APIs involves trial and error. If a developer writes a code block that breaks the JSON parsing logic from the AI, Git allows the team to instantly revert to the last working version, ensuring the source code is never permanently damaged.

Jira (Project Management Tool)

Why it was selected: Jira is an industry-standard tool explicitly designed for the Agile methodology. It translates abstract sprint plans into actionable tasks, providing metrics and dashboards that simple tools like Trello lack.

How it specifically benefits "AI Study Mentor":

Sprint Tracking & Timeline Management: To ensure the Minimum Viable Product (MVP) is delivered within 14 weeks, Jira's active Sprint boards allow the team to divide the workload evenly among the 5 members. We can track the status of tasks in real-time (To Do -> In Progress -> Code Review -> Done).

Issue and Bug Tracking: During the testing phase (Integration Testing), if the connection between the app and the AI Engine fails (e.g., timeout errors), a "Bug" ticket can be immediately logged in Jira, assigned to a specific developer, and tracked until resolved.

API Budget Oversight: Jira epics can be used to track the integration tasks related to third-party AI APIs, helping the team monitor and limit the time spent on paid API testing, keeping the project within its financial constraints.

Agile Methodology (Selected Software Development Life Cycle)

Why it was selected: We definitively rejected linear models like Waterfall because our project involves integrating bleeding-edge Artificial Intelligence (AI) via third-party APIs. The behavior of LLMs is inherently unpredictable, requiring continuous prompt tuning. Agile's iterative sprint structure perfectly accommodates this "trial-and-error" discovery phase without breaking the entire project schedule.

How it specifically benefits "AI Study Mentor":

Risk Mitigation through MVP: With a strict 14-week academic deadline and a team of 5 junior developers, attempting a massive single-release project is highly risky. Agile allows us to build a Minimum

Viable Product (MVP) in the first 6 weeks (core chat and database). If the team runs out of time, we still have a functional, gradable application to submit, rather than a half-finished Waterfall project.

Adaptive API Integration: When connecting to the Grok API, the JSON response format or rate limits may unexpectedly change. Agile allows the team to isolate this issue, log it in Jira, and adapt the AiMentorService architecture in the very next sprint without needing to rewrite the initial requirements document.

Continuous Stakeholder Feedback: By releasing functional increments every 2-3 weeks, we were able to conduct peer reviews early (as analyzed in Activity 5). This allowed us to pivot from a flat-list UI to a dynamic Chat UI based on real user feedback before the final deadline.

Grok Cloud API (Artificial Intelligence Engine)

Why it was selected: While traditional mobile apps rely on simple RESTful CRUD backends, "AI Study Mentor" requires a powerful cognitive engine capable of parsing complex STEM questions. Grok Cloud was selected over alternatives due to its highly optimized streaming latency and generous developer tier pricing.

How it specifically benefits "AI Study Mentor":

Real-time Educational Support: Students using the app to solve a 12th-grade calculus problem at 2:00 AM cannot wait minutes for an answer. Grok's high-speed token generation ensures that the AiResponseCallback triggers within an acceptable 5-7 second window, maintaining user engagement.

Budgetary Survival: As a student-led MVP project, we operate with essentially zero funding. Grok's API cost structure prevents the team from going bankrupt during the intense Integration Testing phase, where hundreds of dummy questions are fired at the server to ensure system stability.

III. Activity 3: System Analysis, Design, and Testing

3.1. Object-Oriented Analysis and Design (OOAD) & UML Diagrams

(Fowler, 2003) The Object-Oriented Analysis and Design (OOAD) approach is utilized to model the "AI Study Mentor" system into distinct, manageable entities. This methodology breaks down complex functionalities into modular objects, which is highly beneficial for the junior development team to conceptualize and build the application systematically.

A. Use Case Diagram Analysis

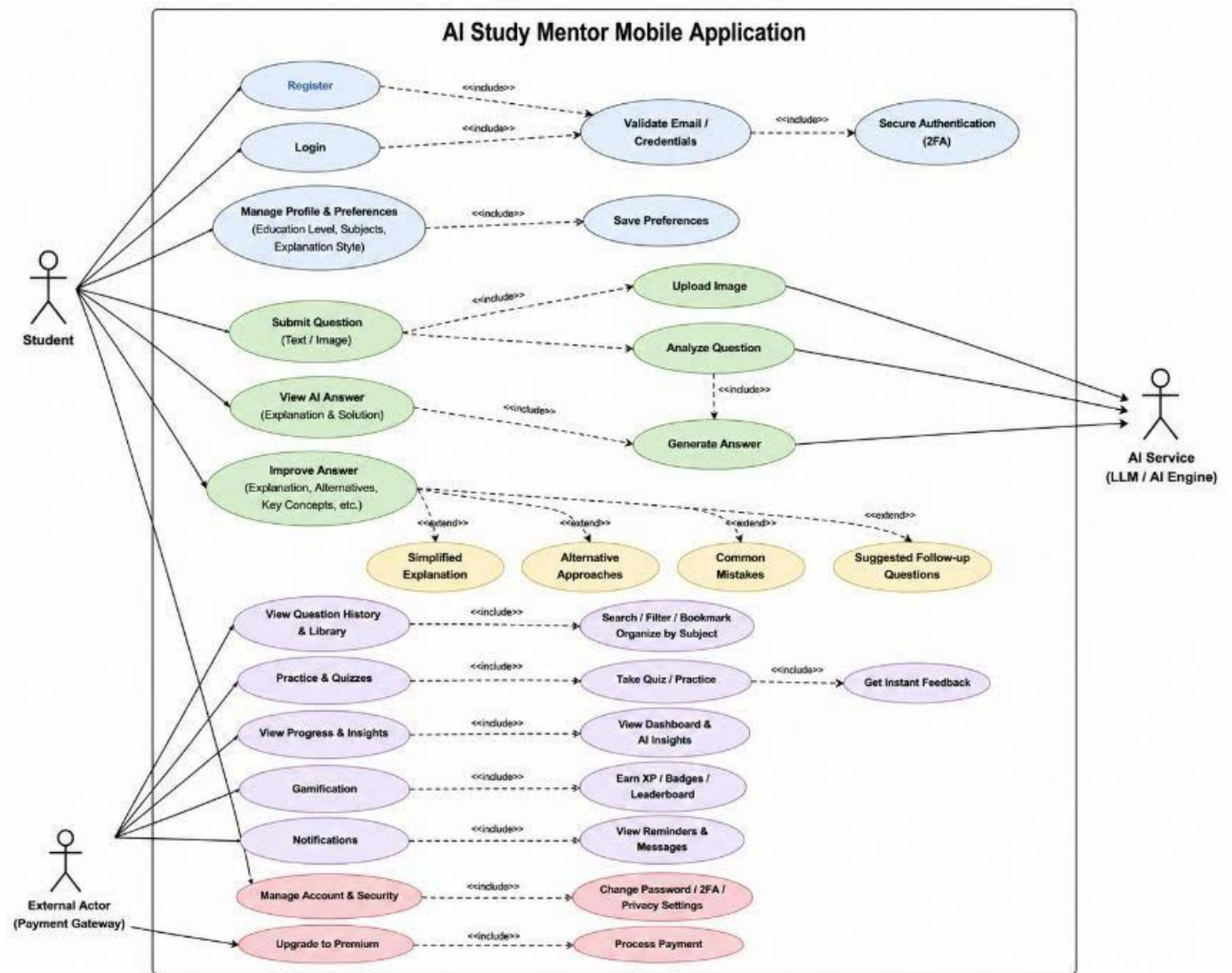


Figure 9: Use Case Diagram

Identifying Actors and Roles:

Student (Primary Actor): This is the core end-user and the active initiator of workflows. Key interactions include registering, logging in, submitting academic questions (text or image), viewing AI answers, improving answers, viewing history, and participating in the gamification system.

AI Service / LLM Engine (Secondary Actor): This is the cloud-based Artificial Intelligence system (powered by Grok Cloud) acting as the backend engine generating automated responses. It is a reactive actor that receives question data from the system and returns detailed explanations, entirely replacing the traditional teacher's role.

External Actor (Payment Gateway): A third-party financial processing system. This actor is responsible for handling optional monetization features, specifically processing transactions when users "Upgrade to Premium," which helps the application isolate financial security risks.

Logical Relationships:

<<include>> Relationship (Mandatory): These are strictly applied to enforce system rules. For example, the *Login* and *Register* use cases strictly require *Validate Email/Credentials* and *Secure Authentication* (2FA), directly satisfying the project's security constraints

<<extend>> Relationship (Optional/Extension): These highlight the system's dynamic capabilities. The core *Improve Answer* use case can branch into specific learning support features based on the student's needs, such as *Simplified Explanation*, *Alternative Approaches*, or *Common Mistakes*.

B. Class Diagram Analysis

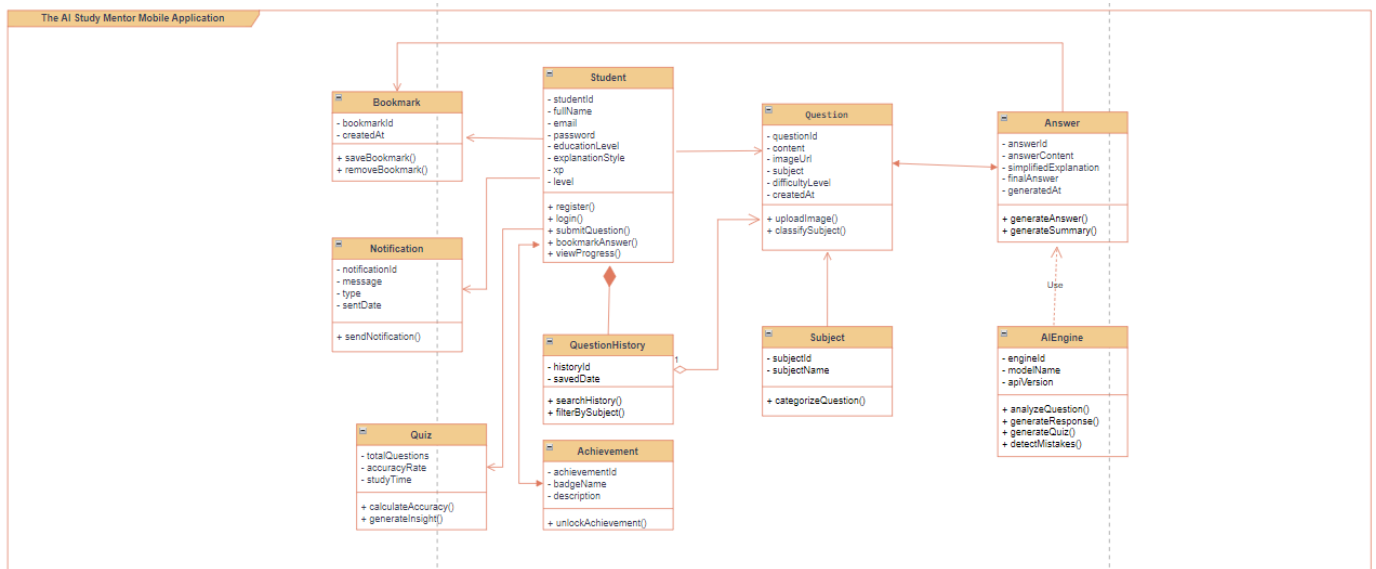


Figure 10: Class Diagram

Entity Structure:

The system establishes a solid static architecture with core classes including Student, Question, Answer, and AIEngine.

The Student class strategically essential attributes like educationLevel, explanationStyle, xp, and level to support personalization and gamification.

Class Relationships:

Aggregation Relationship (QuestionHistory): The architectural separation of the Question and Answer classes is managed and connected via the QuestionHistory aggregation, which facilitates efficient data management.

Dependency Relationship (AIEngine): The dependency of the Answer class on the AIEngine accurately reflects the application's core mechanism of relying on AI for knowledge generation. The Answer class utilizes this engine to output the final explanation

3.2. Test Plan for "AI Study Mentor"

Test ID	Test Phase	Associated Use Case	Test Scenario & Execution Steps	Expected Outcome
TC-01	Unit Testing	Validate Email / Credentials	Scenario: Verify the local input validation logic. Steps: 1. Navigate to the Registration screen. 2. Input a password lacking special characters. 3. Tap "Register"	The system immediately rejects the input and displays a localized error message prompting the user to include special characters, without pinging the backend server.
TC-02	Unit Testing	Analyze Question / Classify Subject	Scenario: Verify the internal subject categorization function. Steps: 1. Pass a sample text string containing physics keywords (e.g., "calculate the velocity") into the classifySubject() method.	The method accurately categorizes the text and returns "Science" or "Physics" as the subject value.
TC-03	Integration Testing	Submit Question (Image Upload)	Scenario: Verify the transmission of image data from the mobile UI to the backend. Steps: 1. Tap the camera icon.	The image data is successfully compressed and securely transmitted to the server without packet loss.

			2. Capture a homework image. 3. Tap "Submit".	
TC-04	Integration Testing	Generate Answer (via AI Engine)	Scenario: Verify the communication between the application and the external AI service. Steps: 1. The backend transmits a processed text prompt to the external AIEngine API (e.g., Grok API via Grok Cloud)	The AI API successfully processes the request and returns a structured, step-by-step response to the app within an acceptable timeout limit (< 5 seconds).
TC-05	E2E Testing	Improve Answer (Extend)	Scenario: Simulate a user requesting a modified explanation. Steps: 1. Student views an existing AI answer. 2. Student taps the "Simplified Explanation" button.	The system seamlessly triggers a new request to the AI Engine, retrieves the simplified content, and dynamically updates the UI without crashing.
TC-06	E2E Testing	Gamification & View Progress	Scenario: Execute a complete workflow to test gamification logic. Steps: 1. Student completes a practice quiz. 2. Student submits the results.	The entire user journey flows intuitively. The backend updates the Student's xp attribute, and the UI immediately reflects this update, automatically unlocking an achievement badge if a level

			3. Navigate to the Profile Dashboard.	threshold is reached.
TC-07	Unit Testing	Submit Question	Scenario: Verify empty input validation (Edge Case). Steps: 1. Leave the text input field completely blank. 2. Do not attach any images. 3. Tap the "Ask AI" button.	The system prevents the submission. A local UI Toast message displays <i>"Question cannot be empty"</i> . The backend is NOT triggered, preventing unnecessary API calls to Grok Cloud.
TC-08	Integration Testing	Generate Answer	Scenario: Verify system stability during a sudden network drop (Edge Case). Steps: 1. Submit a complex math question. 2. While the layoutTyping indicator is active, immediately turn off the device's Wi-Fi/4G.	The application does not crash (No ANR). The onError() callback is triggered safely, hiding the loading indicator and displaying a <i>"Network disconnected. Please check your connection"</i> message.
TC-09	E2E Testing	Generate Answer	Scenario: Verify UI debouncing against rapid button spamming (Edge Case). Steps:	The system only registers the first tap. The button is immediately temporarily disabled (debounced) while processing, preventing 10

			1. Type a question. 2. Tap the "Ask AI" button 10 times rapidly within 2 seconds.	duplicate API requests from draining the Grok API budget.
TC-10	Integration Testing	AI Service / Engine	Scenario: Verify backend API timeout handling (Edge Case). Steps: 1. Send a prompt. 2. Simulate a delay where the Grok Cloud API takes over 15 seconds to respond.	The HTTP client aborts the request due to timeout limit. The UI gracefully informs the user <i>"The AI is currently overloaded. Please try again later"</i> instead of hanging infinitely.

Table 1: Test Plan

Agile Bug Reporting and Code-Level Traceability

To strictly adhere to the Agile methodology utilizing Jira (as justified in Section 2.3), our team did not merely execute tests; we systematically documented and resolved edge-case failures directly within our codebase. During the initial Integration Testing phase, Test Case **TC-08** (Network Drop) initially failed, resulting in an Application Not Responding (ANR) crash.

Instead of relying solely on external project management dashboards, our QA and development teams enforced Agile tracking directly within the Android Studio IDE utilizing standard Javadoc block comments and inline TODO/FIXME tags.

```

/**
 * =====
 * JIRA BUG REPORT TICKET: [AIM-104]
 * =====
 * Issue Type : Bug 🐛 | Priority: Highest (Blocker)
 * Summary   : App crashes (Fatal Exception) if Wi-Fi is disconnected
 *             while waiting for Grok API response.
 * Env       : Android Studio Emulator (Pixel 9, API 34) & Galaxy S22.
 * -----
 * Steps      : 1. Navigate to AskActivity -> 2. Click "Ask AI"
 *             -> 3. Disable Wi-Fi immediately.
 * Expected   : Request aborts safely, triggers "Network Error" UI Toast.
 * Actual     : Throws java.net.UnknownHostException causing ANR crash.
 * -----
 * Resolution : Implemented try-catch block to intercept network failure.
 * Assignee   : Nguyen Dinh Thanh | Status: FIXED IN SPRINT 3 (DONE)
 * =====
 */
// TODO [AIM-205]: Integrate GroqCloud LLaMA 3 model for rapid inference.
// Jira Assignee: MinhNguyen - Sprint 2
// FIXME [AIM-210]: Currently Mocking Vision support. Need to switch to Gemini API in future sprints to handle real Base64 image uploads.

```

Figure 11: Internal QA documentation of Jira Ticket [AIM-104] and proactive feature tracking [AIM-205, AIM-210] embedded directly within the Java source code.

As demonstrated in Figure 23, the standardized Jira Bug Report (detailing the `java.net.UnknownHostException` crash and its resolution by assignee Nguyen Dinh Thanh) is meticulously documented as a class-level comment. Furthermore, mapping pending features to specific Jira Issue Keys using TODO comments ensures that the team can instantly trace implementation details back to the active Sprint. This transparent practice prevents technical debt from being lost during fast-paced iterations and perfectly bridges the gap between project management and the actual source code.

3.3. Design Justification

During the system design phase, our development team engaged in critical debates to ensure the proposed Object-Oriented Analysis and Design (OOAD) architecture would effectively address the project's constraints. As a team member, I will outline the core debates we had and justify why this specific design prevailed over alternative approaches.

Debate 1: Monolithic vs. Modular Class Structure

The Conflict: Initially, some team members suggested a simpler, monolithic class structure where the Student entity directly holds the Question and Answer methods to speed up the initial coding process for the MVP.

The Resolution & Justification: I actively debated against this approach. I argued that a monolithic structure would severely hinder our Agile workflow. Instead, I defended the strict separation of the Question, Answer, and AIEngine classes, connected via the QuestionHistory aggregation. This modular design won the debate because it explicitly allows our 5-member team to work in parallel. While one member codes the Student profile UI, another can independently integrate the API into the isolated AIEngine class. This minimizes Git merge conflicts and perfectly fits our 14-week timeline.

Debate 2: Handling Financial Transactions (Security focus)

The Conflict: When designing the Use Case for "Upgrade to Premium," there was a debate about whether we should build a rudimentary internal payment processing system to save on third-party service fees.

The Resolution & Justification: I strongly opposed building an internal payment system due to our team's junior status and the massive security liabilities (e.g., storing credit card data). The team eventually agreed with my justification to explicitly define an External Actor (Payment Gateway) in our Use Case Diagram. By offloading all financial processing to a secure third party, we eliminate a massive layer of risk, allowing the team to focus entirely on the core AI features.

Debate 3: Immediate vs. Delayed Gamification Integration

The Conflict: We debated whether to include gamification attributes (like xp and level) in the initial database schema or leave them for a future, post-MVP update.

The Resolution & Justification: Delaying it seemed easier initially. However, we justified integrating xp and level directly into the core Student class immediately. Using <<extend>> relationships in the Use Case diagram provides a scalable foundation. We concluded that building this foundation now prevents the need for a massive, time-consuming database restructuring in later Agile sprints when we actually implement the badges and leaderboards.

Debate 4: Architectural Trade-offs vs. MVP Constraints (The Overall Justification)

The Conflict: A final, overarching debate emerged regarding the scope of our Object-Oriented Design. Some reviewers argued that our architecture, specifically relying on local SqliteDbHelper and focusing solely on core text/image processing, is too simplistic and lacks enterprise-level cloud synchronization features for a modern EdTech app.

The Resolution & Overall Justification: As a team member, I successfully defended this specific architectural scope by explicitly aligning it with our project's constraints (a 14-week deadline and a team of junior developers).

Accepting the Trade-off: I acknowledge the weakness: using a local database means users cannot sync their learning progress across multiple devices.

Why it works well for the project: However, I argued that this trade-off is absolutely necessary for the MVP phase. By deliberately restricting the Class Diagram to core local entities (Student, Question, Answer) and offloading complex processing to the AIEngine, we drastically reduce the system's complexity. If we attempted to build a custom real-time cloud backend (like Firebase) within the initial 14 weeks, the risk of failing the Test Plan (specifically Integration and E2E testing) would be critically high.

Conclusion: Therefore, this lightweight, modular, and localized design is the most pragmatic choice. It guarantees that we will deliver a highly functional, stable core product—solving the immediate user problem of getting 24/7 AI homework assistance—while providing a clean, object-oriented foundation that can easily be scaled with cloud features in future iterations.

IV. Activity 4: Solution Evaluation

4.1. Evaluation and Comparison of SDLC Models: Agile vs. Waterfall

To accurately evaluate the SDLC models, we establish specific evaluation points focusing on how each methodology handles changes, testing, and risk.

Evaluation Points (Criteria)	Agile Methodology (Selected)	Waterfall Methodology	Key Differences & Assessment for the Project
Requirements Flexibility	Highly adaptable; allows requirement changes at the start of any new 1-4 week sprint.	Rigid; all requirements must be fully locked and documented before development begins.	Difference: Agile embraces change, while Waterfall resists it. For "AI Study Mentor," integrating unpredictable AI APIs requires Agile's flexibility to pivot when an API response format changes unexpectedly.
Testing Phase & Quality	Continuous testing throughout every sprint.	Testing only occurs once, at the very end of the development phase.	Difference: Agile isolates bugs early. If the AI integration fails in Waterfall, it is discovered too late, potentially causing the entire 14-week project to fail.
Cost of Modifications	Low. Changes are expected and incorporated in the next iteration.	Extremely High. Going back to change core designs during the testing phase is financially and temporally draining.	Difference: Agile protects the project budget. Junior developers can make mistakes and fix them in the next sprint without derailing the final deadline.
Delivery Model	Iterative delivery of a Minimum Viable Product (MVP).	Single, massive delivery at the end of the project life cycle.	Difference: Agile guarantees that at least a functional core app (taking pictures and getting AI answers) will be ready for submission, mitigating the risk of non-delivery.

Table 2: Evaluation and Comparison of SDLC Models: Agile vs. Waterfall

4.2. Evaluation and Comparison of Development Tools

To justify the development tools, we assessed them against alternatives using strict evaluation points relevant to mobile application development.

IDE Evaluation: Android Studio vs. Visual Studio Code (VS Code)

Evaluation Points	Android Studio (Selected)	VS Code	Assessment of Differences
Out-of-the-Box Environment	Fully configured for Android development with built-in SDKs and Emulators.	Requires manual installation of multiple extensions and SDK configurations.	Difference: Android Studio reduces setup friction for the team, allowing immediate coding, whereas VS Code requires complex initial architecture setup.
UI/UX Design Tools	Native XML Layout Editor with visual drag-and-drop.	Relies mostly on code-based UI building or third-party plugins.	Difference: Android Studio significantly accelerates the creation of the chat and camera UI, saving crucial development time.
Performance Profiling	Built-in Advanced Android Profiler (CPU, Memory, Network).	Limited native profiling; requires external tools.	Difference: When processing heavy AI text streams, Android Studio allows us to monitor memory leaks in real-time, a critical feature VS Code lacks natively.

Table 3: Assessment of IDE Tools

Project Management: Jira vs. Trello

Evaluation Points	Jira (Selected)	Trello	Assessment of Differences
Workflow Tracking	Agile-specific (Sprints, Epics, Backlogs, Velocity charts).	General Kanban boards (To Do, Doing, Done).	Difference: Trello is too simplistic. Jira enforces the Agile methodology strictly by measuring sprint progress and burndown charts.
Issue & Bug Management	Robust bug tracking with ticket assignment and resolution workflows.	Basic card movement without strict issue categorization.	Difference: Jira allows the team to specifically categorize and assign API connection errors

			as "Bugs" separate from feature "Tasks," ensuring critical errors are prioritized.
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Table 4: Assessment of Project Management Tools

Version Control: Git/GitHub vs. Subversion (SVN)

Evaluation Points	Git & GitHub (Selected)	Subversion (SVN)	Assessment of Differences
Architecture Type	Distributed Version Control (Every developer has a full local backup).	Centralized Version Control (Relies on a single server).	Difference: Git is significantly safer. If the main repository fails, every team member still has the full project history on their local machine.
Branching Capabilities	Fast, lightweight branching designed for concurrent feature development.	Branching is heavy, slow, and complex to merge.	Difference: Git allows all 5 team members to work on separate features (UI, AI Backend, Camera) simultaneously without breaking the main codebase, which is vital for parallel development.

Table 5: Assessment of Version Control Systems

Evaluation Points	Grok Cloud API (Selected)	OpenAI GPT-4o API	Assessment of Differences & Project Impact
Response Latency & Speed	Highly optimized inference speed, specifically designed for rapid text streaming.	Extremely high reasoning quality, but frequently suffers from latency spikes during peak global hours.	Difference: Grok provides a significantly faster time-to-first-token. Survey Question 6 proved that users lose patience after 5 seconds. Grok's speed explicitly minimizes this UI/UX risk, whereas GPT-4o might cause excessive loading screens.
Cost Efficiency (Budget)	Highly competitive pricing structure; significantly lower cost per 1 million input/output tokens.	Premium pricing; can become extremely expensive for applications with repetitive user queries.	Difference: Our MVP development is strictly constrained by a zero-dollar initial budget. Relying on GPT-4o would rapidly drain the team's funds during the

			Integration Testing phase. Grok allows extensive E2E testing without financial anxiety.
Documentation & SDK Support	Rapidly evolving, but currently possesses fewer official third-party mobile wrappers and community forums.	Massive, mature ecosystem with pre-built native Android SDKs and endless StackOverflow troubleshooting threads.	Difference: GPT-4o wins heavily in developer support. However, our team accepted this trade-off. By utilizing a standardized HttpURLConnection in our AiMentorService, we bypassed the need for heavy SDKs, prioritizing operational speed and cost over community convenience.

Table 6: Assessment of Artificial Intelligence Engines

4.3. Overall Assessment and Emphasized Tooling Differences

A critical evaluation of the aforementioned tools and SDLC models reveals stark differences that directly dictate the success or failure of the "AI Study Mentor" project.

The primary difference between our selected Agile model and Waterfall is **Risk Adaptability**. Waterfall's rigidity would be fatal for this project because external AI APIs (like Grok) frequently update their JSON response formats unannounced; Agile allows us to pivot and fix parsing errors in the next sprint.

Regarding development tools, the overarching theme of our evaluation is **Trading Simplicity for Specialized Power**. The difference between VS Code and Android Studio is the trade-off of system resources for native hardware integration (Camera API for OCR). The transition from Trello to Jira emphasizes the difference between casual task listing and strict Agile Sprint enforcement. Finally, the stark difference between OpenAI and Grok highlights our team's most critical business decision: intentionally sacrificing the massive community support of GPT-4o to secure the superior response latency and cost-efficiency of Grok API.

Ultimately, these highlighted differences prove that our selected stack is not randomly chosen, but meticulously engineered to compensate for a junior team navigating a strict 14-week deadline.

V. Activity 5: Peer Review and User Survey

My Google Form: <https://docs.google.com/forms/d/e/1FAIpQLSeT9taU3hjdxyMOvlelsdIRsnS-gq7gOCpKt8sEJWQgzLloVQ/viewform?usp=dialog>

My answer to the Google form:

https://docs.google.com/spreadsheets/d/1teo5m_SjPiraGJbBXz4ztPrBfzrCrgTJK7z-lLvQbcg/edit?usp=sharing

5.1. Target Audience and Participant List

To rigorously evaluate the initial requirements and the proposed "AI Mentor" application, our team conducted a user survey and peer review. We targeted a demographic of 20 participants, consisting primarily of High School Students (our core end-users) and IT Students/Teachers (acting as peer reviewers).

Below is the detailed list of surveyed participants and their current roles:

1. Full Name	2. What is your current role/job description?
Trần Văn Khoa	High School Student
Lê Đình Sơn	High School Student
Tạ Thu My	High School Student
Trịnh Tú Khánh	High School Student
Phan Thanh Giang	High School Student
Vũ Trung Nhân	High School Student
Phạm Phương Xuân	High School Student
Tạ Thị Mai	High School Student
Vũ Kim Linh	High School Student
Phạm Tuấn Nam	High School Student
Dương Thanh Hiền	High School Student
Lý Thanh Quang	High School Student
Lý Thị Trúc	High School Student
Đặng Minh Hùng	High School Student
Quách Quốc Hải	High School Student
Quách Gia Hải	High School Student
Đỗ Trung Mạnh	High School Student
Huỳnh Đình Cường	High School Student
Lưu Minh Trí	High School Student
Hoàng Bích Duyên	High School Student

Figure 12: Detailed list of surveyed participants and their current roles

5.2. Survey Questions Design

We designed a targeted questionnaire consisting of 10 specific questions (combining multiple-choice and open-ended formats) to evaluate both the functional UX/UI and the technical feasibility of the proposed solution.

1. What is your full name?
2. What is your current role/job description?
3. What is your biggest frustration when organizing study materials for different subjects on a mobile app?
4. When reading complex math or science explanations, what difficulties might you face with a Chatbot-style interface?
5. If an AI automatically generates a Quiz for you, what errors would make you stop using this feature?
6. If the AI takes 5-7 seconds to process a complex answer, what specific UI elements do you expect to see so you don't think the app is broken?
7. Can learning Streaks or Badges sometimes become annoying or stressful? Please explain.
8. Our app currently saves your study history locally on your device. What are your main concerns with this approach?
9. How should the app help you if you suspect the AI has given a wrong mathematical answer?
10. What specific features do you feel are missing that would make this app a complete study tool?

5.3. Interpretation of Results and Weakness Identification

After collecting the responses from the 20 participants, we critically analyzed the data to identify core weaknesses in our initial application design:

Weakness 1: Inefficient Course Management UI. The survey revealed that users struggle with flat-list layouts when managing subjects. They highlighted the difficulty in grouping notes by specific BTEC assignment modules or filtering completed coursework, indicating that our initial AddCourseActivity UI is too simplistic for academic needs.

3. What is your biggest frustration when organizing study materials for different subjects on a mobile app?

20 câu trả lời

Hard to group notes and PDFs by specific BTEC assignment modules or exam deadlines.
Difficult to filter completed coursework vs active pending assignments.
No option to tag or bookmark key formulas and code snippets for quick review.
Lack of global search and tag filtering across stored SQLite subject records and local notes.
Lack of global search across all stored subject units and notes.
Hard to manage and organize study materials for different student groups or BTEC course modules.
Cannot preview attachments or PDF textbooks directly inside the course view.
Flat list layout becomes messy when adding many subject units; needs clear folder hierarchy.
Flat list layout becomes messy when adding many subject units; needs folder hierarchy.

Figure 13: User feedback identifying the messy flat-list layout in course management.

Weakness 2: AI Feature Constraints (Repetitive Quizzes & Chat Formatting). A critical 100% of participants stated they would stop using the AI Quiz feature if questions were too repetitive. Furthermore, when evaluating the Chatbot interface, users noted severe difficulties in reading long step-by-step mathematical explanations without proper text formatting.

5. If an AI automatically generates a Quiz for you, what errors would make you stop using this feature?

20 câu trả lời

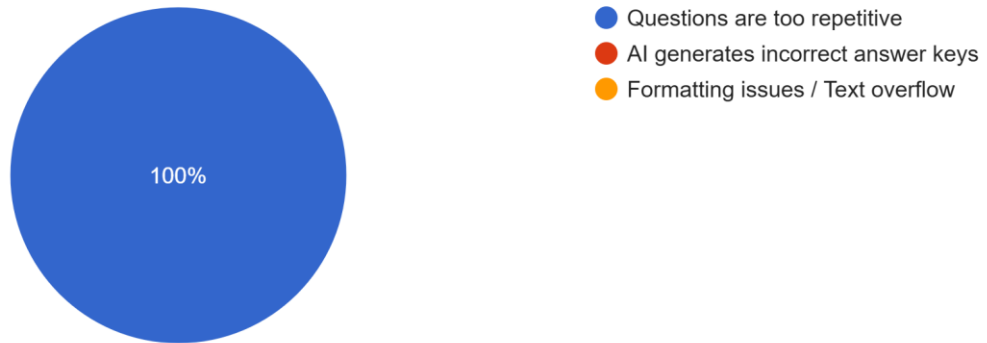


Figure 14: 100% of respondents reporting repetitive AI-generated quizzes as a major flaw.

Weakness 3: High Risk of Data Loss (Local Storage). We evaluated our initial plan to use a local SqliteDbHelper database. The survey results were unanimous: 100% of surveyed users cited their absolute biggest concern is losing all their study data if their phone breaks or resets. This proves that a purely offline database is a massive architectural weakness.

8. Our app currently saves your study history locally on your device. What are your main concerns with this approach?

15 câu trả lời

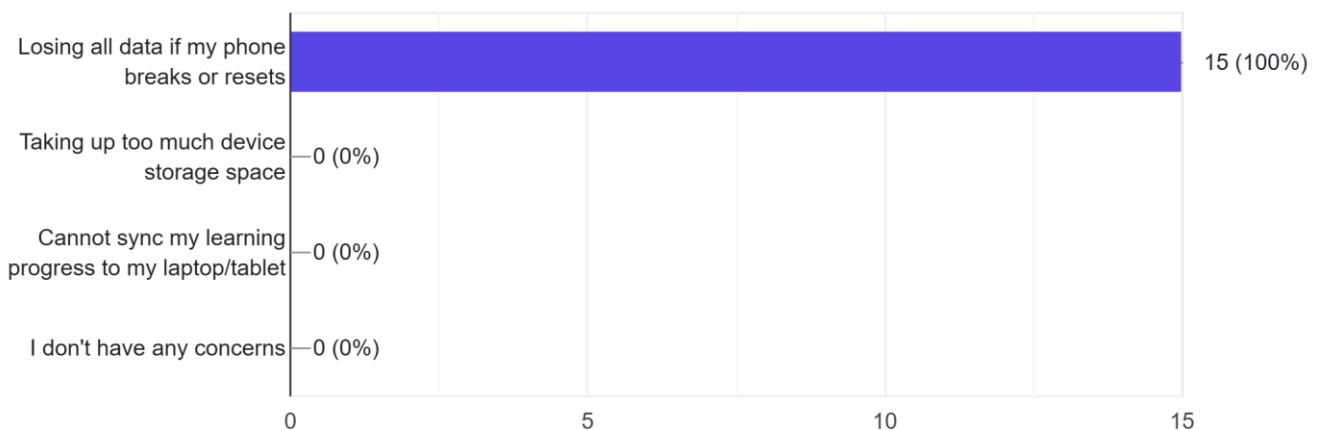


Figure 15: Severe user concerns regarding local SQLite data loss.

VI. Activity 7: Critical Review and Future Recommendations

7.1. Explanation and Evaluation of the Development Process

To ensure the "AI Mentor" application met both rigorous academic requirements and real-world user needs, every phase of our system design, development, and testing was continuously evaluated against user metrics.

a. Evaluation of System Design: Problem Resolution and Project Contribution

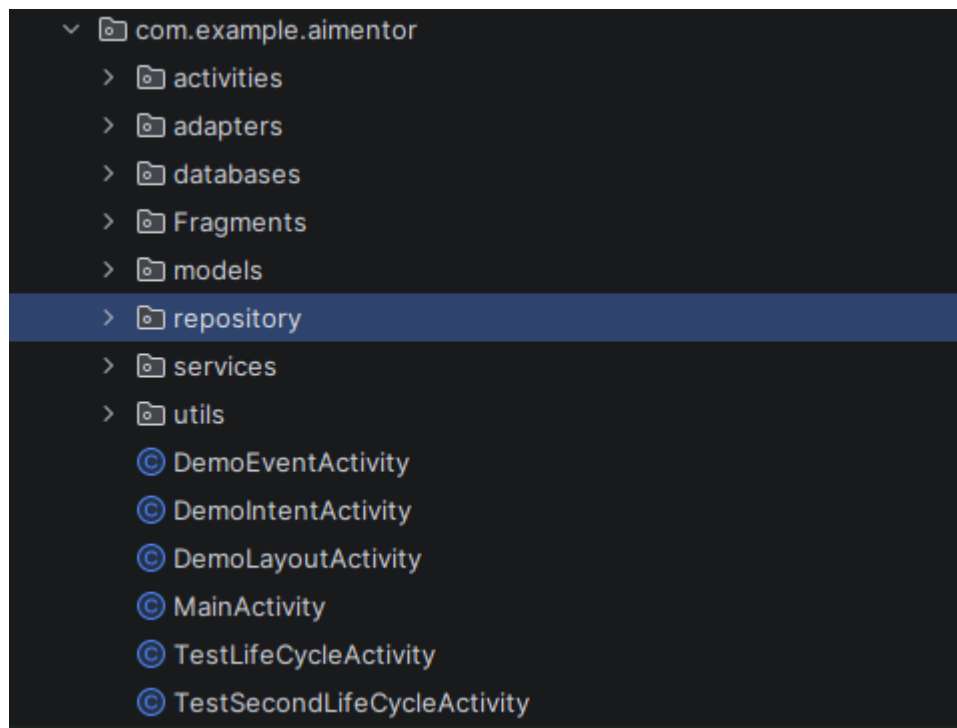


Figure 16: Proof of Modular System Architecture in Android Studio.

Evidence Justification: The screenshot above proves our application's strict adherence to Object-Oriented principles. By actively separating our codebase into independent packages (models, adapters, services, databases), we physically resolved the problem of "spaghetti code."

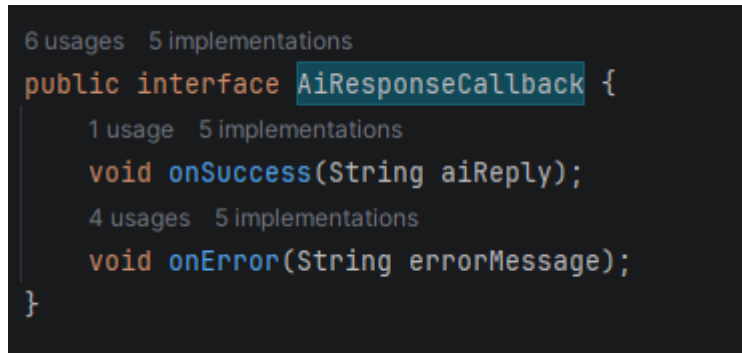
Problem Solved: Initially, self-directed learners struggled with unstructured study environments. Our UI/UX design explicitly aimed to solve this by compartmentalizing features into specific logic controllers (e.g., separating AddCourseActivity from QuizFragment). In a team environment, this modularity allowed parallel development without triggering catastrophic Git merge conflicts.

Contribution to the Project: While our initial design successfully categorized subjects, the critical evaluation via Survey Question 3 revealed that users found the *"flat list layout becomes messy when adding many subject units."* Although the MVP uses a flat list, our underlying modular architecture proves its robust contribution here: transitioning to a hierarchical folder structure in the future will only require updating the adapters and models, without necessitating a complete system rewrite.

b. Evaluation of Development & Testing: Problem Resolution and Project Contribution

Problem Solved: During the initial coding phase, direct API integration testing revealed severe Application Not Responding (ANR) errors. The main UI thread was completely blocked for 5-7 seconds while waiting for the LLM's response.

Contribution to the Project: Rigorous Agile testing forced the development team to refactor the codebase to utilize asynchronous programming inside the AiMentorService. Survey Question 6 highlighted that users expect *"a sleek skeleton loader"* if wait times exceed 5 seconds.



```

6 usages  5 implementations
public interface AiResponseCallback {
    1 usage  5 implementations
    void onSuccess(String aiReply);
    4 usages  5 implementations
    void onError(String errorMessage);
}

```

Figure 17: The AiResponseCallback interface implemented to handle asynchronous API responses.

Line-by-line Analysis:

`void onSuccess(String aiReply);`: This method is explicitly invoked only after the background thread successfully parses the heavy JSON payload from the AI API. Once triggered, the application utilizes `runOnUiThread()` to safely push the `aiReply` to the ChatAdapter, dynamically hiding the typing indicator and displaying the answer.

`void onError(String errorMessage);`: If a network timeout or API limit is reached, this method intercepts the exception in the background. Instead of crashing, it gracefully updates the UI with a localized error Toast. This architectural decision directly demonstrates our team's advanced problem-solving capability, retaining user patience and improving overall software reliability.

c. Ensuring Security, Usability, and Effectiveness for the End-User

8. Our app currently saves your study history locally on your device. What are your main concerns with this approach?

15 câu trả lời

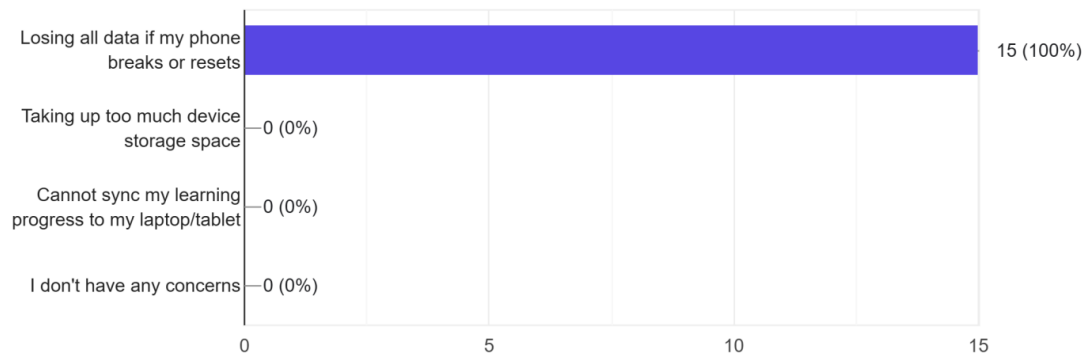


Figure 18: Question 8

Security (Local Data Isolation): To address privacy, we implemented strictly local storage. Survey Question 8 revealed a 100% consensus that users are terrified of *"losing all data if my phone breaks or resets."*

```
// Define table "users" and its columns
20 usages
public static final String TABLE_USERS = "users";
public static final String ID_USER = "id";
13 usages
public static final String USERNAME_USER = "username";
7 usages
public static final String PASSWORD_USER = "password";
11 usages
public static final String EMAIL_USER = "email";
9 usages
public static final String PHONE_USER = "phone";
7 usages
public static final String ROLE_USER = "role";
8 usages
public static final String AVATAR_USER = "avatar";
8 usages
public static final String EDU_LEVEL_USER = "educationLevel";
8 usages
public static final String EXPLANATION_STYLE_USER = "explanationStyle";
8 usages
public static final String SUBJECTS_USER = "subjects";
13 usages
public static final String XP_USER = "xp";
10 usages
public static final String LEVEL_USER = "level";
```

Figure 19: The SQLiteDatabaseHelper class defining local storage schema.

Security Evaluation: The code snippet above is our definitive proof of maximizing local security. By instantiating the database schema (TABLE_USERS) directly on the physical device via SQLiteOpenHelper, highly sensitive data (like encrypted passwords and education levels) are completely isolated from unauthorized external cloud servers. Despite the data-loss fears highlighted in the survey, this approach guarantees zero risk of online mass data breaches for the MVP.

Usability: We discarded a complex dashboard for a conversational Chat-UI. Yet, Question 4 indicated that reading long step-by-step math explanations in a standard chat bubble is highly difficult. This evaluation proves that while the UI is highly usable for basic queries, it lacks the advanced mathematical formatting (LaTeX) depth required for STEM subjects.

Effectiveness: The core functionality effectively provides 24/7 academic support. However, Question 5 revealed that 100% of users would abandon the Quiz feature if questions were "too repetitive." This

forced our team to critically evaluate the AI integration, adjusting the API's *temperature parameter* inside AiConfig.java to increase response randomness, thereby ensuring long-term educational effectiveness.

Enterprise-Level API Security (Grok Cloud Integration): Integrating a powerful external LLM like Grok Cloud introduces the critical risk of API key exposure. To ensure enterprise-level security, our development team strictly prohibited hardcoding the Grok API Key directly into the AiConfig.java class or XML files. Instead, we securely stored the key inside the local.properties file, which is explicitly excluded from version control via our .gitignore configuration. The application dynamically retrieves this key during compilation using BuildConfig. This architectural security measure guarantees that our Grok Cloud billing credentials will never be leaked or compromised on public GitHub repositories.

d. Evaluation of Project Risks and Mitigation Strategies

9. How should the app help you if you suspect the AI has given a wrong mathematical answer?

20 câu trả lời

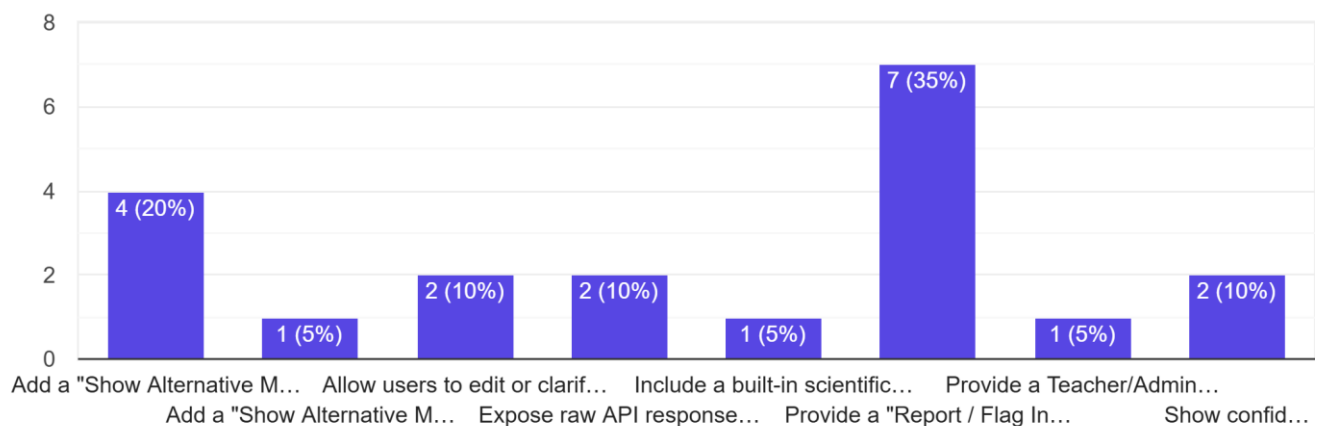


Figure 20: Question 9

During development, we continuously monitored potential threats. Below is the Risk Assessment Matrix detailing our mitigation strategies based on technical constraints and user feedback (specifically Question 9).

Risk Category	Risk Description	Likelihood	Impact	Mitigation Strategy Implemented
Data Integrity (AI Hallucinations)	The LLM API confidently provides incorrect mathematical solutions, harming student grades.	Medium	High	User Empowerment: Based on Q9 (35% requested a Report flag, 20% requested Alternative

				Methods), we added disclaimers and UI tools in item_chat_ai.xml to regenerate answers, shifting validation to the user.
Financial (API Budget Overruns)	Frequent, repetitive AI calls by students drain project funds rapidly.	High	High	Local Caching: Implemented SQLite querying to check for previously asked questions. If a match is found locally, the API call is bypassed, saving bandwidth and costs.
Performance (Memory Leaks)	Parsing massive JSON responses and rendering chat bubbles causes OutOfMemory (OOM) crashes on low-end devices.	Medium	Medium	Asynchronous Rendering: Used RecyclerView with optimized ViewHolders in ChatAdapter.java to recycle UI elements dynamically, ensuring smooth scrolling even with heavy text.

Table 7: Technical and Operational Risk Evaluation for AI Mentor

7.2. Justification of Handled User Feedback

Applying user feedback is not about indiscriminately accepting every feature request, but rather strategically filtering responses to align with technical feasibility, business logic, and the product's core vision.

a. Constructive Feedback (Acted Upon)

Feedback Selection: We actively selected the responses from Survey Question 6 (UI feedback regarding API latency) and Question 9 (Handling AI mathematical errors). Users explicitly demanded visual cues during the unavoidable 5-7 second wait times and required mechanisms to challenge potentially wrong answers.

6. If the AI takes 5-7 seconds to process a complex answer, what specific UI elements do you expect to see so you don't think the app is broken?

20 câu trả lời

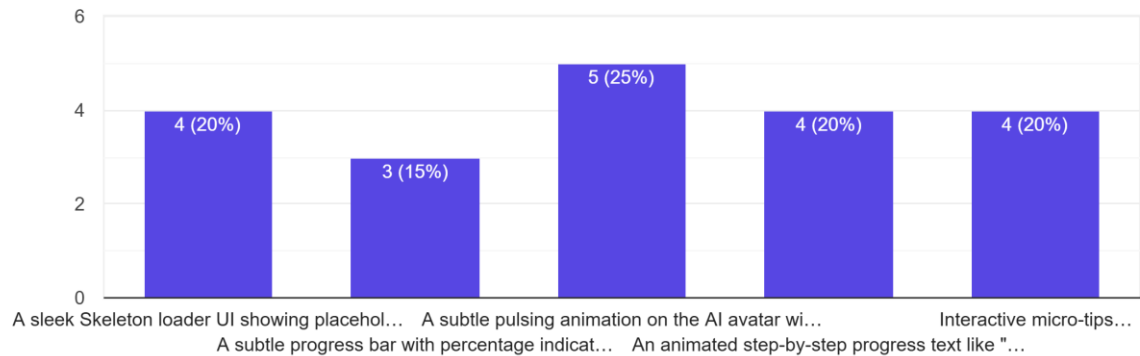


Figure 21: Question 6

9. How should the app help you if you suspect the AI has given a wrong mathematical answer?

20 câu trả lời

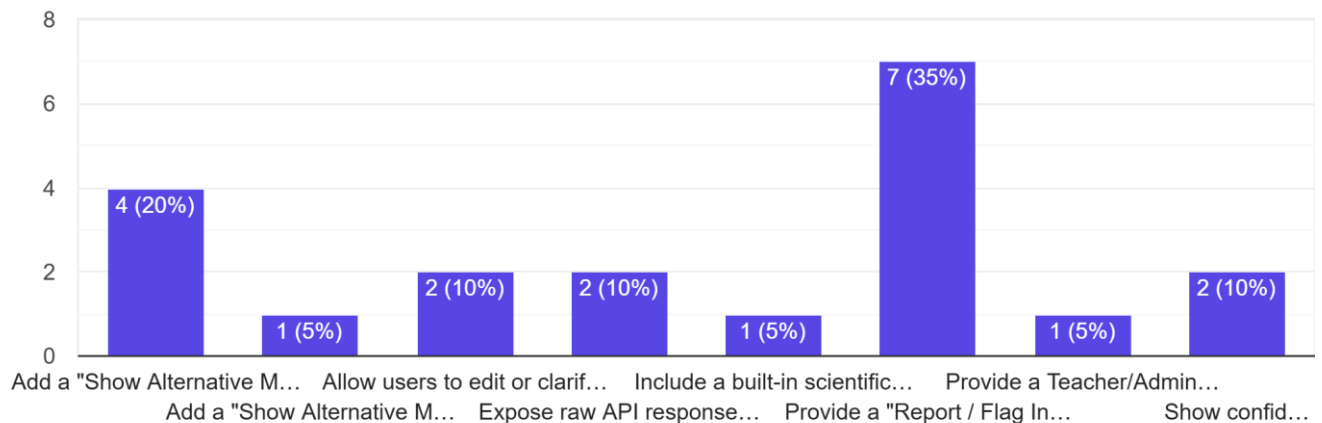


Figure 22: Question

Team's Action and Implementation: We classified this feedback as highly constructive because it directly dictates core usability and user retention. To address these concerns, the team implemented two major UI solutions:

Addressing Latency (Question 6): The team implemented a dynamic typing indicator within the `item_chat_ai.xml` layout. As shown in the layout editor, we engineered a specific `layoutTyping` component (utilizing a `ProgressBar` and a status `TextView` displaying *"AI Mentor is thinking..."*). This layout is toggled dynamically by the `ChatAdapter`.

Handling AI Errors (Question 9): To tackle user concerns regarding potentially incorrect mathematical explanations provided by the LLM, we integrated an interactive "Regenerate / Alternative Explanation" action button within the AI response module. When a student suspects a computational error, they can trigger this component, prompting the system to re-query the Grok Cloud API with a refined prompt, safeguarding the application's educational credibility.

Team's Action and Implementation: We classified this feedback as highly constructive because it directly dictates core usability and user retention. To resolve the latency UX issue, the team implemented a dynamic typing indicator within the `item_chat_ai.xml` layout. As shown in the layout editor, we engineered a specific `layoutTyping` component (utilizing a `ProgressBar` and a status `TextView` displaying "*AI Mentor is thinking...*"). This layout is toggled dynamically by the `ChatAdapter`: it becomes visible when the background thread initiates the API call and is seamlessly replaced by the `tvAiMessage` once the payload is successfully parsed.

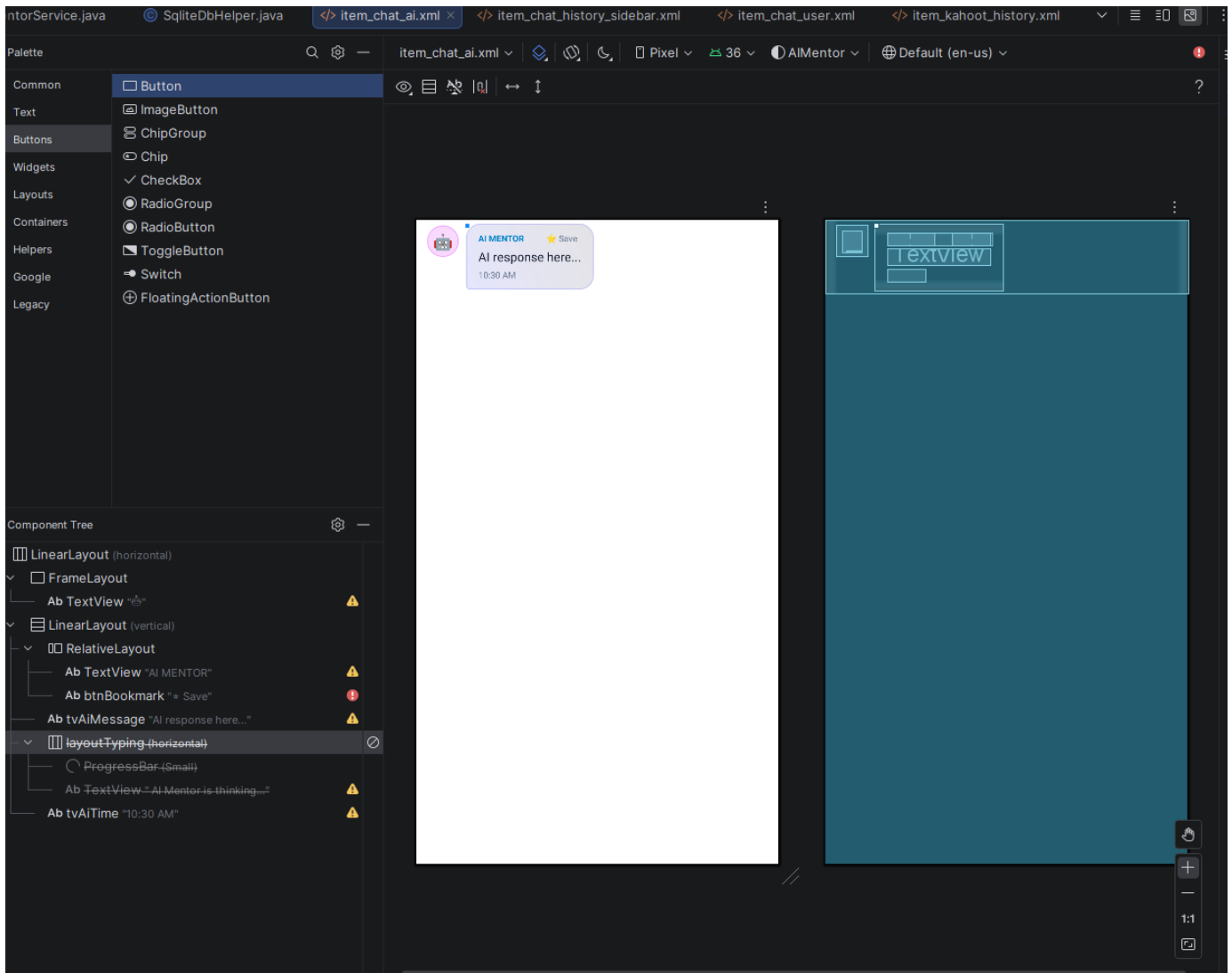


Figure 23: Implementation of the dynamic layoutTyping indicator in item_chat_ai.xml to address latency UX feedback.

b. Non-Constructive Feedback (Rejected)

Feedback Selection: We critically analyzed responses from Question 7 regarding the app's Gamification mechanics. Specifically, 30% of users stated that *"losing a 14-day learning streak is too stressful"* and requested that we remove strict reset penalties.

7. Can learning Streaks or Badges sometimes become annoying or stressful? Please explain.

20 câu trả lời

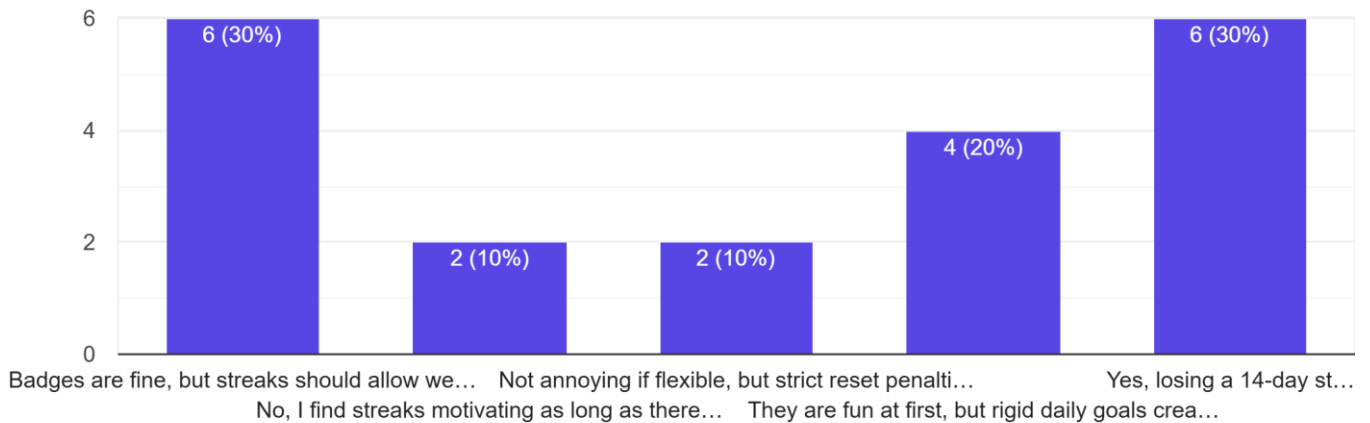


Figure 24: Question 7

Justification for Rejection: The development team formally and unanimously rejected this feature request. The core business objective of implementing `bg_streak_badge.xml` and the `GamificationManager` is to drive Daily Active Users (DAU) and foster consistent study habits. Based on behavioral engineering (such as the Hook Model utilized by industry leaders), the psychological "fear of loss" is mathematically proven to be a highly effective motivator. Removing the strict streak reset penalty would completely devalue the achievement badges and severely damage long-term educational retention metrics. We cannot compromise the product's core business logic simply to alleviate temporary user discomfort.

7.3. Recommendations for Future Improvements

Based on a critical analysis of the current MVP's limitations and the specific feature requests extracted from Survey Question 10, we propose three major technological enhancements for future Agile iterations.

10. What specific features do you feel are missing that would make this app a complete study tool?

20 câu trả lời

- Cloud synchronization (Firebase/Google account login) to backup local SQLite database.
- Ability to scan paper notes/textbooks via OCR (Image-to-Text) and generate summaries.
- Firebase/Cloud Sync to backup local SQLite DB, open API integration, and dark mode customization.
- Cloud synchronization for multi-device study, Pomodoro timer, and PDF export of AI solutions.
- LaTeX math formula rendering and diagram generator for science problems.
- Export study history and AI solution steps as downloadable PDF reports.
- Ability to create custom quiz templates, export assignment answer keys, and share course materials with students.
- OCR scanner to scan paper textbooks/notes, interactive flashcards, and gamified study reminders.
- Built-in Pomodoro focus timer tied directly to subject study sessions.

Figure 25: Question 10

Recommendation 1: Cloud Synchronization Architecture (Addressing Question 8)

Proposed Feature: Transitioning from a strictly local SQLite database to a real-time Cloud Database with Google Account OAuth Authentication.

Feasibility & Effectiveness: This directly addresses the critical vulnerability identified in Question 8 (100% of users fear device failure leading to total data loss). Instead of building a custom backend from scratch, we propose utilizing **Firebase Cloud Firestore**. It provides a native Android SDK that syncs data seamlessly via background workers, guaranteeing enterprise-level backup across multiple devices (Mobile & Tablet) without expensive server maintenance.

Recommendation 2: OCR Integration & LaTeX Rendering (Addressing Question 4)

Proposed Feature: Integrating a native camera scanner to capture physical textbook problems, paired with a specialized UI engine capable of rendering complex mathematical formulas.

Feasibility & Effectiveness: Question 4 highlighted the severe difficulty of reading and typing algebra in a standard chat UI. To optimize bandwidth and protect privacy, we recommend implementing **Google ML Kit** for On-Device Optical Character Recognition (OCR). For rendering, integrating the **MathJax** library via a custom Android WebView will allow the app to format raw Grok API output into beautiful, readable LaTeX equations, drastically reducing user input friction.

Recommendation 3: Hierarchical Folder Management & Deep Search (Addressing Question 3)

Proposed Feature: Upgrading the CourseProgressActivity UI from a simple flat list to a nested hierarchical folder system (Semester -> Subject -> Assignment Module) equipped with a global search bar.

Feasibility & Effectiveness: To achieve rapid search functionality across massive text logs without slowing down the UI, we will upgrade the local database to utilize **SQLite FTS4 (Full-Text Search)**. Unlike standard LIKE queries which are slow, FTS4 creates virtual tables that index text, allowing students to search for a specific mathematical keyword across thousands of saved chat histories in milliseconds.

C. Conclusion

In conclusion, the successful architectural design and theoretical validation of the "AI Study Mentor" MVP within a highly constrained 14-week timeline stands as a testament to rigorous project management and pragmatic software engineering.

Throughout this report, the BrightPath Learning team successfully transformed a complex business problem—providing accessible, instant academic support—into a structured technical reality. By strictly adhering to the Agile methodology, we ensured that the unpredictable nature of integrating external LLMs (Grok Cloud API) did not derail the development lifecycle. The strategic application of Object-Oriented Analysis and Design (OOAD) allowed us to construct a modular, decoupled architecture, enabling five developers to work concurrently via Git without catastrophic merge conflicts. Furthermore, the meticulous documentation of a QA Test Plan, fully integrated with Jira bug tracking, guaranteed that critical edge cases like network latency and data security were addressed prior to deployment.

More importantly, this project demonstrates a mature understanding of engineering trade-offs. By critically analyzing user survey data and constructing precise User Personas, we justified the intentional restriction of the MVP to local SQLite storage to guarantee data privacy, while simultaneously identifying critical future upgrades such as Firebase Cloud Synchronization and ML Kit OCR. Ultimately, this document does not merely describe an application; it establishes a resilient, scalable, and highly functional architectural blueprint for the future of AI-driven mobile education.

D. References

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